

Why Human Beliefs Are Neither Fully Rational Nor Irrational

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Introduction

Humans believe in a wide variety of things - from information passed down by ancestors to both visible and invisible phenomena happening in the present, as well as imagined possibilities that have not yet happened. Beliefs are ubiquitous, shape how we view the world, process new information, interact with others, and influence our experiences.

Being rational means forming beliefs based on logic and evidence, rather than emotion or impulse, and updating them when contrary evidence becomes available. However, there are many beliefs people hold that don't follow this rational way of thinking. People across the world believe in all kinds of things that are not based on logic or evidence - superstitions, paranormal beliefs, conspiracy theories, religious beliefs, etc, yet those who believe in them provide explanations and evidence that seem logical to them. Also, people who once held such beliefs and later lost faith don't agree that they were pretending (Van Leeuwen, 2023, as cited in Sommer, 2026).

If these beliefs seem rational to some and irrational to others, why do people hold them? What role do these beliefs play? In what ways are these beliefs formed, why do they persist, and why do they change? Why do people sometimes hold inconsistent beliefs? How do limitations of human cognition influence the formation and updating of these beliefs? Are these beliefs rational or irrational?

There are multiple, interacting factors that shape how people come to hold particular beliefs, often yielding different answers to these questions. In the following sections, I first define what

beliefs are and outline different types of beliefs. I then examine the role beliefs play in providing instrumental, cognitive, and hedonic value, how the availability of evidence influences belief formation, and why beliefs persist over time. I also address why people sometimes hold inconsistent beliefs, and argue that limitations in both human cognition and the environment together constrain belief formation and updating, making beliefs rationally bounded rather than simply rational or irrational.

What Is a Belief?

Beliefs are information about ourselves and the world stored in our minds in the form of mental representations with varying levels of confidence (Wedgwood, 2012 as cited in Oktar, 2025). A dispositionalist view of belief suggests that to believe something means to have some behaviour associated with the belief. Similar to dispositionalism, interpretationism focuses on observable behaviour. According to functionalism, beliefs are not identified by how they are represented, rather they are identified by the functions they serve, such as guiding behaviour (Schwitzgebel, 2024). However, beliefs do not always lead to behaviour (Fishbein & Ajzen, 1975, as cited in Sommer, 2025). In many cases, a single belief does not guide behaviour, rather, multiple beliefs are required, and the correlation between beliefs and behaviour is often minimal (Sommer, 2025).

Types of Beliefs

Beliefs are often divided into categories based on their function, acquisition, representation in the mind, and other criteria. Most of these categories appear to divide beliefs into those that are considered rational because they are supported by evidence and those that are considered not rational because they are either subjective or lack direct evidence.

Testable beliefs are formed based on evidence, either through direct observation (e.g., the sky is blue) or indirect information from experts or authorities (e.g., global warming is increasing). They are objective and can be verified. Distal beliefs, on the other hand, are subjective, cannot be verified, and are often held for instrumental reasons (e.g., to feel good) (Abelson, 1986, as cited in Sommer, 2026).

Based on their representation in the mind, beliefs can be categorized into intuitive or reflective. Intuitive beliefs are processed quickly and inferred automatically through perception (e.g., when we see someone crying, we infer that they are sad) without deliberate thought. Reflective beliefs, by contrast, are meta-representations about beliefs and are formed after considering reasons for holding them (e.g., the Earth is round because the scientists have demonstrated it). Ideological, religious, moral, and political beliefs are not intuitive, as they can be held without complete understanding or full endorsement (Sperber, 1996, 1997, as cited in Sommer, 2026).

Another categorization distinguishes between factual beliefs and imaginations. Ideologies, religious, and political beliefs are often classified as imaginations. Factual beliefs are defined by

four principles. First, they are involuntary, meaning we do not choose to believe them (e.g., we don't choose to believe that the grass is green, it happens automatically), whereas imagination is voluntary (e.g., we choose to believe whether God exists or not). Second, factual beliefs guide behaviour and are not compartmentalized. Imagination, in contrast, involves holding two representations at once - one about reality and the other about pretense, resulting in compartmentalization. It also doesn't guide behaviour (e.g., when imagining a banana as a phone, people don't expect it to function as one). Third, inferences are made based on factual beliefs (e.g., when someone imagines a ball going up, they imagine the ball coming down based on their knowledge about gravity). Fourth, factual beliefs are vulnerable to evidence and should be updated when counterevidence is available (Van Leeuwen, 2023, as cited in Sommer, 2026).

Imagination, distal and reflective beliefs are not always formed based on evidence, but they do play an important role in our daily functioning.

The Role of Beliefs

Often, we don't pay attention to how and why we form our beliefs. When and why do we search for more information to support our beliefs, or stop engaging with information that contradicts them? What purpose do holding some beliefs serve?

If the rational way to form or change a belief is to consider the evidence carefully and revise accordingly, many beliefs people hold don't follow this ideal Bayesian rule of updating. Instead, information is sought or ignored to maximize instrumental, hedonic, and cognitive needs, often at the cost of accuracy (Sharot, 2020, 2023).

Instrumental Value

Information has instrumental value if it can be used to achieve a goal or obtain a reward. It functions as a performance-enhancing device, for example, people may seek feedback to improve performance while also avoiding it to maintain their performance. In some cases, information about a candidate is deliberately avoided before an interview to promote impartiality and fairness (Hertwig, 2016). When information holds positive instrumental value, people prefer to know more about it. Conversely, when it could lead to negative outcomes, it is often ignored, using information strategically (Sharot, 2023).

Sometimes, beliefs can change neurobiology without conscious awareness, as seen in the placebo effect, where the belief that a treatment will work leads to measurable neurobiological changes (Porot & Mandelbaum, 2021). Beliefs in luck and extra sensory perception serve an instrumental function by reducing both performance-related and existential anxiety associated with uncertainty (Žeželj, 2019).

Similarly, people hold accuracy-dependent beliefs when there is a potential gain or a fear of punishment (Sharot, 2023). For example, beliefs about whether one has prepared sufficiently for an exam. In this case, accurate beliefs lead to better exam outcomes, whereas inaccurate ones may result in failure.

Hedonic Value

Information also has hedonic value, as it helps in emotion regulation. People seek information that increases positive affect and avoid information that increases negative affect. At times, people avoid information to induce positive affect such as surprise, and seek information to reduce negative emotions that come with anxiety due to uncertainty (Sharot, 2020; Hertwig, 2016). Research on affective forecasting suggests that people's ability to predict their future affective reactions is skewed, making them overestimate the duration of negative emotional states, which can lead to unnecessary information avoidance (Gilbert & Wilson, 2007, as cited in Hertwig, 2016).

Religious beliefs have been associated with greater life satisfaction, positive emotions, optimism, lowering symptoms of anxiety and depression, and a stronger sense of meaning in life. These positive effects may be mediated by adaptive emotion regulation strategies, such as cognitive reappraisal and acceptance, as well as reduced rumination (Vishkin, 2019). Conspiracy theories often emerge in response to crises that generate fear and anxiety due to uncertainty. These emotional states create a need for sense-making, which leads individuals to believe conspiracy explanations to reduce uncertainty. Over time, these beliefs are transmitted across generations as representations of historical events (Van Prooijen, 2017).

People hold accuracy-independent beliefs for rewards, social acceptance or to maintain a positive internal state (e.g., feeling good, optimism) . When their identity is connected to a group, people may resist evidence and rationalize falsehoods, fake news and conspiracy theories. (Van Bavel, 2019, as cited in Sharot, 2023).

Cognitive Value

Information provides cognitive value by building internal models of the world and the self. As confirmation bias suggests, people seek information that reinforces their preexisting beliefs and preserves their mental models. People try to maintain coherence between their mental models and reality by interpreting reality in ways that fit their existing model, rather than changing the model itself. This involves selectively seeking information that strengthens their understanding of the world while ignoring information that threatens it (Sharot, 2020).

Given the vast amount of information available today, deliberate avoidance also functions as an information-management tool to sustain cognitive resources such as attention. Heuristics are often used instead of gathering all possible information to reduce mental effort (Hertwig, 2016).

The way information is evaluated based on its usefulness rather than its accuracy helps explain why some beliefs are maintained even in the absence of strong evidence, especially beliefs that do not align with reasoning or behaviour, and are epistemically suspect such as conspiracies, pseudoscientific beliefs, paranormal and superstitious beliefs, or unsupported self-related beliefs.

How Instrumental, Hedonic, and Cognitive Values Shape Information Use

When new information is encountered, weights are assigned to instrumental, hedonic, and cognitive utilities to obtain an overall information value, which influences whether the information is sought or avoided. Illusion of control, unrealistic optimism, impact bias, illusion

of knowledge, overconfidence, and other biases shape how these weights are assigned. When instrumental utility is held constant, people tend to prefer options with higher hedonic utility over cognitive utility. Additionally, when losses outweigh gain, people are more likely to avoid information (Sharot, 2020).

However, most of these calculations happen in the background, outside conscious awareness. The instrumental, hedonic, and cognitive values highlight the purposes beliefs serve and how they are processed by the mind. Given the vast amount of information and evidence available in the world, an important question is how this evidence shapes belief formation.

The Role of Evidence in Belief Formation and Updating

Although some beliefs provide hedonic benefits initially, they turn out to be more damaging in the long run. People continue to believe in conspiracy theories about past events even when they no longer experience uncertainty about those events, and these beliefs further reinforce feelings of fear, anxiety, and existential threat (Van Prooijen, 2017). Belief formation and updating are not solely driven by personal or internal factors, the environment that provides information also plays a crucial role. When people encounter something unexpected, it generates a learning prediction error, which in turn linearly predicts belief change (Vlasceanu, Morais & Coman, 2021, as cited in Orticio, 2021). This effect is stronger when uncertainty about a statement is high and when observed social prevalence is higher than expected. As a result, people update their beliefs more in response to social prevalence information when they are uncertain than when they are certain (Orticio, 2021).

Sommer (2026) introduced the idea of locality of evidence. Evidence about a belief is highly local when all relevant information is readily available. For example, perceptual evidence such as the sky is blue. In such cases, belief formation becomes straightforward, and individuals tend to independently converge on the same conclusion, resulting in objective agreement. In contrast, non-local evidence is dispersed, with information not available all at once, requiring time and effort to gather the information. For example, evidence about climate change comes from multiple sources, such as melting glaciers, rising temperatures, and disappearing coral reefs. At the other extreme end, where the evidence is highly non-local (also referred to as isotropic), almost any piece of information could be relevant to the belief. This makes it difficult to converge on a single conclusion, leading to subjective opinions. Religious, moral, and political beliefs, as well as ideologies are formed under such conditions, which contributes to persistent disagreement, as different perspectives can all appear valid.

For beliefs to form reliably, a high proportion of local evidence is required. When local evidence is limited, people form beliefs through uncertainty absorption (March and Simon, 1958, as cited in Sommer, 2026). There is so much information available, often making it impractical to gather all the relevant details before forming a belief. In situations where uncertainty is high due to a high proportion of non-local evidence, people rely on experts or authorities. When beliefs are formed through uncertainty absorption, individuals focus more on the source of the evidence (who is providing it) rather than the content (how the belief was formed or what evidence supports it). As a result, when beliefs are adopted from a common or authoritative source without independent verification, counterevidence may have little impact on belief change because

people may assume that additional supporting evidence exists that they have not yet encountered but that the source might be aware of (Sommer, 2026).

This helps explain why many political, religious, and ideological debates rarely reach consensus. The high proportion of non-local evidence allows for multiple interpretations to seem plausible, and because beliefs are also shaped by personal motivations, individuals may perceive their own views as more rational than those of others. These environmental factors not only shape belief formation but also help explain why belief revision can become immune to opposing evidence.

Belief formation and updating are influenced by both internal processes and external structures. It is not solely a matter of the mind or the world, but a combination of both that shapes how beliefs are formed and maintained.

Multiple Pathways Influencing Belief Persistence

The environmental factors such as locality of evidence and uncertainty absorption not only influence how beliefs are formed but also how they persist in the face of opposing evidence. Persisting in belief despite counterevidence has been studied under various terms, including confirmation bias, closed-mindedness, cognitive rigidity and conservatism in updating (Oktar, 2025).

In domains with non-local evidence, disagreement often functions as disconfirming evidence, which is a form of higher-order evidence. When people encounter such evidence, it can lead to four different outcomes. First, they may agree with the opposition and shift their belief in favor

of the counterevidence presented, especially when people are uncertain about the statement (Orticio, 2021) and when the evidence comes from aggregated opinions (Oktar, Lombrozo, & Griffiths, 2024 as cited in Oktar, 2025). Second, when faced with disagreement, they may refrain from making a judgement about whether their belief should be updated (Langenhoff et al., 2023 as cited in Oktar, 2025). Third, after considering the disconfirming evidence, they may choose to stick with their existing belief, especially when the evidence comes from an unreliable source (Kominsky et al., 2016 as cited in Oktar, 2025). Fourth, they may become more polarized and strengthen their existing belief, particularly when they perceive the counterevidence as misleading or manipulative (Cook & Lewandowsky, 2016 as cited in Oktar, 2025).

There are several reasons why a belief may persist despite encountering opposing evidence. The concepts of locality of evidence and uncertainty absorption explain how the availability of information influences our beliefs. The interaction between these factors can be better understood through the different routes by which beliefs persist. The Paths to Persistence model (Oktar, 2025) proposes four pathways that contribute to belief persistence, and the interaction among the pathways makes it difficult to disentangle why belief change is challenging, especially when evidence is non-local. The 4 paths are as follows:

The **Informational Path** involves how the evidence is processed. When people perceive others' informational processes as unreliable, they form judgements about inferior evidence and inferior processing of information. Individuals who believe their own views reflect reality (naive realism) tend to see their evidence as more objective, assign less weight to peer input, consider them uninformed and lacking evidence, leading to belief persistence (Lieberman et al, 2012, as cited in

Oktar, 2025). People may also attribute disagreement to lower intelligence or bias in others (Oktar, 2025), which further reinforces their initial beliefs and reduces openness to alternative perspectives.

The **Ontological Path** involves judgments about subjectivity and unknowability. When people consider issues to be subjective, they are less likely to seek social consensus information and more likely to discount others' views (Egan, 2010 as cited in Oktar, 2025). In cases of uncertainty absorption, where beliefs are formed through non-local evidence via trusted sources, beliefs may persist because individuals assume relevant evidence exists somewhere that they have not yet accessed (Sommer, 2026). This is particularly common in religious beliefs, making them continue with their existing beliefs. Ontological confusion is a common feature of superstitious beliefs, beliefs in luck, paranormal phenomena, and ESP beliefs (Žeželj, 2019).

The **Functional Path** involves both interpersonal and intrapersonal drivers of belief. Intrapersonally, beliefs may be maintained because they help people feel good by reducing uncertainty about the world. Interpersonally, belief change may threaten social belonging or status in groups that require adherence to specific moral, political, or religious views, leading to persistence for social and functional benefits (Oktar, 2025).

The **Computational Path** involves constraints in representations and reasoning. Similar to religious beliefs, when people form scientific beliefs through experts (uncertainty absorption), they inaccurately represent these concepts, leading to inconsistencies and persistence. People also misrepresent others' beliefs, as seen in cases of pluralistic ignorance (Fields & Schuman,

1976, p. 427 as cited in Oktar, 2025), and in false consensus effect, people assume others think similarly and continue behaviours without questioning them (Mullen et al., 1985 as cited in Oktar, 2025).

There may be more than these four pathways, with multiple interactions among them, functioning as higher-order mechanisms underlying belief formation and persistence. Based on these pathways, religious beliefs are associated with ontological path, scientific beliefs are associated with the informational path, and moral and political beliefs are distributed more evenly across all pathways (Oktar, 2025).

While these pathways explain why beliefs persist in the face of opposing evidence, they do not guarantee that beliefs remain internally consistent and coherent across situations. The same constraints that lead to persistence such as reliance on limited information, source-based reasoning, and reduced evaluation of evidence can also result in fragmented and inconsistent beliefs.

Why Do People Hold Inconsistent Beliefs?

Despite the persistence of beliefs, humans often hold inconsistent ones. We often rely on the availability heuristic, using what comes to mind to make decisions or reason about a situation. For example, people may believe airplane crashes are more frequent than car crashes due to their frequent representation in the news. Sommer (2025) shows that humans favour consistency, yet the absence of accessible information in a given moment can lead to inconsistent beliefs, while simultaneous access to relevant information reduces such inconsistencies. In their study, asking

people the following 3 questions in the order [1, 2, 3] led many to answer “No” to “Do whales have hair?”, likely because they did not recall that whales are mammals, and that all mammals have hair. However, when the order was reversed to [3, 2, 1], many people answered “Yes” to “Do whales have hair?”.

1. Do whales have hair?
2. Are whales mammals?
3. Do all mammals have hair?

Participants generated inconsistent answers more often in the [1, 2, 3] order but were also more likely to recognize the inconsistency and revised their beliefs toward consistency, compared to when the order was [3, 2, 1].

If memory were a fully consistent web of connections across various memories, then new information would continuously reshape the entire structure. However, Bendaña & Mandelbaum (2021, as cited in Porot & Mandelbaum, 2021) argue that memory is fragmented, with each fragment representing different aspects of information about the same belief. These fragments are causally isolated, meaning that what happens in one fragment does not necessarily influence another. This provides another explanation for why people hold inconsistent beliefs despite possessing relevant knowledge. They may simply lack access to the necessary information at a given moment.

Another example of memory’s role in inconsistency is the sleeper effect (Kumkale & Albarracín, 2004 as cited in Gilovich, Keltner, Chen & Nisbett, 2018), where messages from unreliable sources don’t influence people initially, but over time, people dissociate the source from the

message, making them more susceptible to persuasion. Fragmentation also helps explain why people behave differently across situations - different contexts could elicit different memories, resulting in variation in behaviour depending on which information is accessible.

These inconsistencies suggest a broader limitation on belief updating, particularly in how information is accessed and integrated over time.

Bounded Rationality

In many real-world settings, evidence is non-local, meaning that information relevant to belief formation is dispersed and difficult to integrate. Along with cognitive limitations such as limited time, knowledge, memory, the ability to forecast future outcomes and other constraints, it becomes difficult to gather and evaluate all relevant information before forming a belief. As a result, people rely on uncertainty absorption and the information available to them at that moment.

Under these conditions, belief updating typically does not follow Bayes' rule, in which prior knowledge and new information are carefully integrated. Instead, changing one's belief involves evaluating existing beliefs against new information that is often incomplete or noisy. Forming and updating beliefs don't always occur based on accurate information; internal factors, such as positive/negative affect (hedonic), internal models of the world (cognitive), and material gains (instrumental) along with external factors, such as information availability and uncertainty play important roles.

Belief change is also shaped by outcomes and rewards. People tend to maintain beliefs associated with positive outcomes and are more likely to change their beliefs when the change itself is associated with a positive outcome (Sharot, 2023). In reinforcement-learning tasks, greater learning is observed following unexpected positive outcomes because they create large reward prediction errors, leading to stronger neural updates (Lefebvre et al., 2017, as cited in Sharot, 2023). Thus, belief updating reflects a combination of uncertainty, environmental cues, and outcome-based learning rather than strict probabilistic integration. This may be why some beliefs appear rational to some, while others may consider them irrational.

Together, these patterns raise a broader question about rationality in belief updating. The concept of bounded rationality suggests that the choices people make are constrained by both the limitations of the mind and the limitations of the world (Simon, 2000). Cognitive constraints include limits on knowledge, memory, and attention, as well as how people process information in light of personal motivations such as instrumental, cognitive, and hedonic values. At the same time, environmental constraints include the unavailability of all information at once (non-local evidence) and uncertainty, shaping how beliefs are formed, maintained, and updated.

Conclusion

People hold beliefs not simply because they are true or false, but because they serve functional roles under uncertainty. Beliefs help people make sense of the world, achieve goals and desires, and regulate emotions, which explains why some beliefs appear rational to those who hold them, yet irrational to others. Beliefs are formed and updated through interactions between existing knowledge, new and often incomplete information, and environmental cues such as social

prevalence. They persist when they continue to provide cognitive, instrumental, or hedonic value, and change when alternative beliefs promise better outcomes. Inconsistent beliefs can arise because, in most cases, information is not readily available or is highly non-local, making people rely on heuristics rather than integrating all evidence simultaneously. Under these conditions, limitations of human cognition and the structure of the environment jointly shape belief formation and updating. As a result, beliefs are neither purely rational nor irrational. Instead, beliefs reflect bounded rationality, where responses to uncertainty, cognitive constraints, and the information available jointly influence belief formation and updating.

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